



Public Transit Route Choice Modeling on Real Networks: Expanded Path Size Logit vs. Nested Recursive Logit

Practicum Report

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Abstract

We compare two route choice estimation approaches Expanded Path Size (EPS) logit and Nested Recursive Logit (NRL) on real transit networks in Sydney and Bengaluru, chosen for contrasting topology (sparse rail spine vs. dense, redundant metro-bus). Using 3,000 synthetic trips per city generated from a known-parameter Path Size Logit process, both models are fit by maximum likelihood on identical choice-set data. EPS recovers line-haul parameters well in Sydney but underestimates walk time, while in Bengaluru it captures walk time accurately but severely underestimates the transfer penalty due to route-overlap over-correction. NRL shows large parameter biases across both networks under current settings, indicating sensitivity to sample size and convergence tuning. The results reveal a trade-off between network redundancy and parameter identifiability, motivating further sensitivity analysis and validation on real trip data.

1 Methodology

The full data processing and model estimation pipeline is illustrated in Figure 1, spanning data collection through EPS/NRL model fitting and evaluation.¹

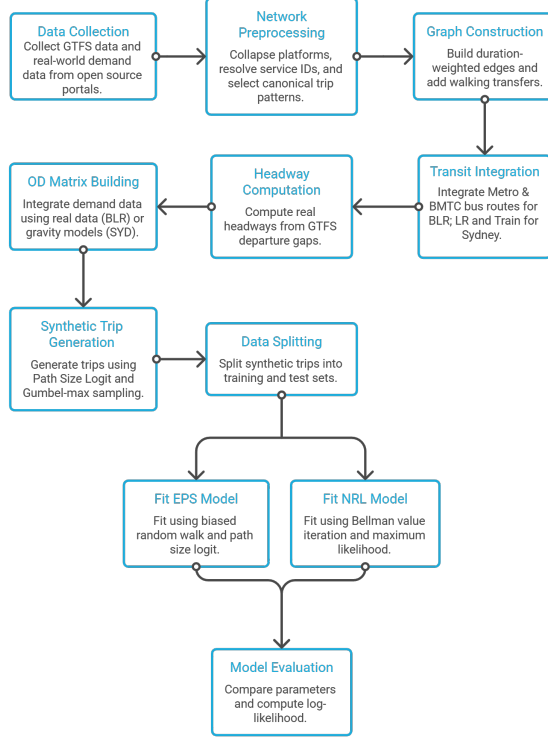


Figure 1: Overview of the data processing and model estimation pipeline.

Comparative implementation of two route-choice estimation methods namely Expanded Path Size Logit (EPS; Frejinger, Bierlaire & Ben-Akiva, 2009) and Nested Recursive Logit (NRL; Mai, Fosgerau & Frejinger, 2015) on real transit networks in Sydney and Bengaluru.

For one origin–destination pair per city (selected as the tractable pair with the richest universal choice set, $|U| \leq 500$), 3000 synthetic trips were generated from a known-parameter Path Size Logit process over every simple path in U . Both estimators are then fit by maximum likelihood on an 80/20 train/test split of the *same* universal-choice-set data (conceptual paradigms are illustrated in Figure 2):

- **EPS** samples R paths per observation via a biased random walk, applies an importance-sampling correction offset, and estimates via L-BFGS-B. The systematic utility V'_{in} for sampled path i includes the correction offset:

$$V'_{in} = -\beta_{ivt} \cdot IVT_i - \beta_{wait} \cdot Wait_i - \beta_{walk} \cdot Walk_i - \beta_{transfer} \cdot N_{trans,i} + \beta_{ps} \ln(EPS_i) + \ln \left(\frac{k_{in}}{R \cdot q_{in}} \right) \quad (1)$$

where k_{in} is the number of times path i was sampled, and q_{in} is the path sampling probability.

¹For Data and Code availability refer here: https://github.com/kartikeya-gaur/PT_route_choice_eps_nrl_mini_project

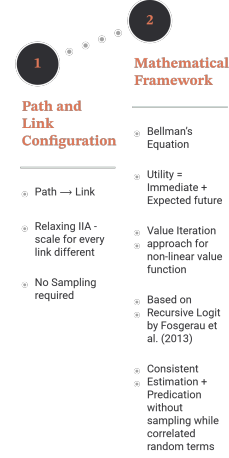
- **NRL** solves the Bellman recursion over link-level value functions for each destination and estimates the model parameters, also via L-BFGS-B. The value function V_i for link i is recursively solved to convergence:

$$V_i = \mu_i \ln \sum_{j \in \Gamma(i)} \exp \left(\frac{u_{ij} + V_j}{\mu_i} \right) \quad (2)$$

where $\Gamma(i)$ represents the successor links of link i , u_{ij} is the instantaneous transition utility from link i to link j , and μ_i is the link-specific scale parameter.

1	Sampling Protocol Initial sampling process
2	Attractive Alternatives Exploring alternative sampling methods
3	Sampling Correction Adjusting sampling due to ASC
4	Stochastic Path Using a biased random walk
5	PSL Ignoring correlation with non-sampled paths
6	EPS Expansion factor based on sampling protocol

(a) EPS path-level sampled choice set framework.



(b) NRL link-level sequential decision recursion.

Figure 2: Methodological estimation paradigms: (a) EPS path-level importance sampling, and (b) NRL link-level sequential Bellman recursion.

2 Study Networks

Sydney and Bengaluru were chosen for contrasting topology (Figure 3, Table 1). Sydney’s network is a near-linear rail spine (T1/T2 largely overlapping from Parramatta through Strathfield into the CBD, with an L2 light-rail branch), where genuine path choice is concentrated in a small CBD interchange cluster. Bengaluru’s network layers two metro lines with three parallel bus routes converging on the Kempegowda/Majestic hub, connected by a much larger number of short walking transfers; redundancy engineered into the network to create richer path choice.

Table 1: Supernetwork topology summary

City	Nodes	Edges	Transfer edges	Walking-transfer stations	Multi-route stations
Sydney	35	94	6	5	10
Bengaluru	36	94	24	17	4

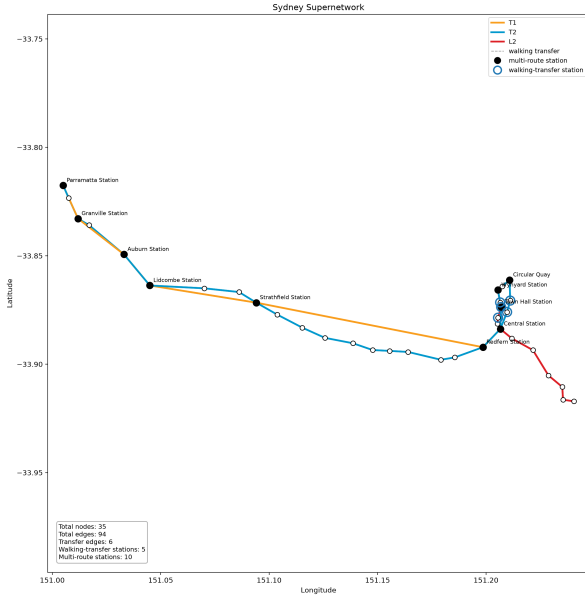
3 Results

3.1 EPS

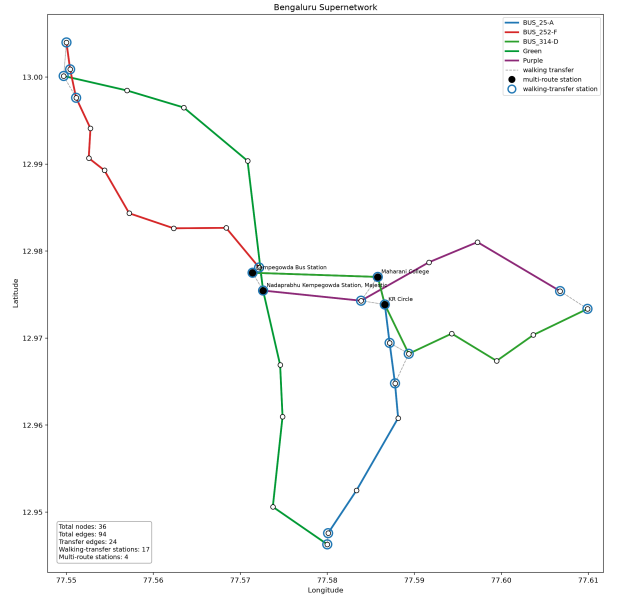
Here, R represents the number of scholastic path generated.

Table 2: EPS parameter estimates and percentage biases across draw counts R (true values: $\beta_{ivt}=0.025$, $\beta_{wait}=0.05$, $\beta_{walk}=0.035$, $\beta_{transfer}=0.4$; obs. retained $\approx 2090-2400$)

City	R	Sampled $ C $	Test LL	Time (s)	$\hat{\beta}_{ivt}$ (% Bias)	$\hat{\beta}_{wait}$ (% Bias)	$\hat{\beta}_{walk}$ (% Bias)	$\hat{\beta}_{transfer}$ (% Bias)
Sydney	10	2.85	-404.0	44.4	0.031 (+24%)	0.050 (0%)	0.011 (-69%)	0.377 (-6%)
Sydney	20	4.64	-672.4	69.7	0.021 (-16%)	0.046 (-8%)	0.006 (-83%)	0.426 (+6%)
Sydney	40	8.30	-989.8	83.4	0.024 (-4%)	0.055 (+10%)	0.015 (-57%)	0.328 (-18%)
Sydney	80	15.40	-1338.9	131.7	0.025 (-1%)	0.049 (-1%)	0.015 (-56%)	0.377 (-6%)
Bengaluru	10	4.64	-646.4	53.5	0.012 (-51%)	0.066 (+33%)	0.022 (-37%)	-0.012 (-103%)
Bengaluru	20	8.00	-954.4	71.3	0.015 (-40%)	0.062 (+24%)	0.033 (-7%)	0.043 (-89%)
Bengaluru	40	13.90	-1293.4	108.3	0.017 (-32%)	0.062 (+24%)	0.034 (-2%)	0.044 (-89%)
Bengaluru	80	23.90	-1620.0	169.3	0.017 (-32%)	0.059 (+18%)	0.036 (+2%)	0.079 (-80%)



(a) Sydney: linear rail spine, path choice concentrated near the CBD.



(b) Bengaluru: metro + parallel bus with dense walking transfers.

Figure 3: Study network topologies. Filled markers are multi-route (interchange) stations; open circles with blue outlines are walking-transfer stations.

3.2 NRL

A random 100-observation training subsample from the 3000 synthetic trips set. Table 3 shows NRL fit on the same underlying data used above

Table 3: NRL parameter recovery, both cities

City	Parameter	True	Estimated	Bias	% Bias
Sydney	β_{ivt}	0.025	0.130	0.105	+421%
Sydney	β_{wait}	0.050	0.094	0.044	+88%
Sydney	β_{walk}	0.035	0.030	-0.005	-15%
Sydney	$\beta_{transfer}$	0.400	-0.389	-0.789	-197%
Bengaluru	β_{ivt}	0.025	0.316	0.291	+1165%
Bengaluru	β_{wait}	0.050	-0.135	-0.185	-370%
Bengaluru	β_{walk}	0.035	0.326	0.291	+832%
Bengaluru	$\beta_{transfer}$	0.400	1.864	1.464	+366%

3.3 Runtime

Runtime for both the city network for both the models

Table 4: Runtime summary, seconds ($R=80$ for EPS)

City	Model	Train (s)	Test (s)	Total (s)
Sydney	EPS	100.9	15.6	116.5
Sydney	NRL	6165.5	4.2	6169.7
Bengaluru	EPS	118.9	20.8	139.7
Bengaluru	NRL	13843.2	8.5	13851.7

4 Insights

4.1 EPS

By Parameters-

- β_{ivt} : Bengaluru’s dense overlapping routes create minor travel-time differences between alternatives, flattening the gradient and suppressing the estimate.
- β_{wait} : Both cities recover the parameter reasonably; Sydney is near-exact while Bengaluru overshoots moderately.

- β_{walk} : Sydney’s short, localized walk segments are under-sampled by the walk protocol, whereas Bengaluru’s frequent, varied walk-heavy routes let $R \geq 40$ fully expose this parameter.
- $\beta_{transfer}$: Bengaluru’s overlapping transit lines push the EPS expansion factor (Φ_j) to over-correct, heavily suppressing the transfer utility gradient the model’s worst-performing estimate overall.

Table 5: Parameter Recovery at $R = 80$ (Bias vs. True Value)

Parameter	Sydney	Bengaluru	Better In
β_{ivt} (in-vehicle time)	−1% (≈ 0.025)	−32% (from $R=40$)	Sydney
β_{wait} (waiting time)	−1% (≈ 0.049)	+18%	Sydney
β_{walk} (walking time)	−56% (persistent)	+2% (from −37% at $R=10$)	Bengaluru
$\beta_{transfer}$ (transfer penalty)	−6% (0.377)	−80% (0.079 vs. true 0.400)	Sydney

By City-

- **Sydney:** Few alternative paths make transfers and line-haul segments distinct and identifiable, while walk segments being short and localized are under-sampled by the random walk.
- **Bengaluru:** Varied and frequently-traversed walk-heavy routes expose walk disutility well, but heavy route overlap makes the EPS expansion factor (Φ_j) over-correct.
- **Overall (Sydney vs. Bengaluru):** Sparse networks favor identification of “big,” discrete choice components over fine-grained, localized ones, whereas dense networks expose fine-grained behavior (walking) at the cost of correlated components (transfers).

5 Way Ahead

- Conduct a proper *sensitivity analysis* for the parameters used in generating synthetic trips.
- Experiment with parameters for the EPS model, including the *number of draws* and the *Kumaraswamy distribution* parameters.
- Develop a deeper understanding of the NRL approach and optimize it for *more run cycles*.
- Establish clarity on the *basis for comparison* between both models, and create scenarios to validate them.
- Building on robust ideas and experimentation, apply the approach to a *real dataset* of trips to generate *behavioural inferences*.